9. ESTIMATION OF CALCIUM (TITRIMETRIC METHOD)

Aim

To estimate the amount of calcium present in the given test solution.

Principle

Calcium is precipitated as Calcium Oxalate with ammonium oxalate. The precipitate is washed with ammonia to remove the chloride ions. The washed precipitate is then made to react with 1N sulphuric acid. The liberated oxalic acid is now estimated by titrating against standardised potassium permanganate. The amount of oxalic acid liberated is proportional to the amount of calcium.

Reagents required

1. **Ammonium Oxalate solution (4%)**

   4g of ammonium oxalate dissolved in 100ml of distilled water.

2. **Ammonia Solution (2%)**

   2ml of ammonia of specific gravity 0.88 is diluted to 100ml with distilled water.

3. **Potassium Permanganate(0.1N)**

   This is prepared by dissolving 3.16 g of potassium permanganate in 1 litre of distilled water.

4. **Standard oxalic acid solution (0.1N)**

   It is prepared by dissolving 630mg of oxalic acid in 100ml of distilled water.

5. **Sulphuric acid(1N)**

6. **Calcium oxalate solution 100mg of calcium oxalate dissolved in 100ml of distilled water**

Procedure

Standardisation of Potassium Permanganate (Titration I)

10ml of oxalic acid is pipetted out into a clean conical flask and 10ml of dilute sulphuric acid is added and heated at 60˚C . It is titrated against potassium permanganate in the burette. The end point is the appearance of pale permanent pink colour. Titration is repeated for concordant values.

Estimation of Calcium Oxalate (Titration II)

10ml of calcium oxalate solution is pipetted out into a conical flask followed by the addtion of 10ml of 1N sulphuric acid and warmed for solubilisation. This solution is now titrated against potassium permanganate and the volume consumed is noted. The end point is the appearance of pale permanent pink colour. Titration is repeated for concordant values.

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